Binary Relation

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1 Binary Relation (二元關係)

I Binary relation

i Binary relation or correspondence

A binary relation or correspondence R over sets X and Y is a subset of $X \times Y$. The set X is called the domain or set of departure of R, and the set Y the codomain or set of destination of R. In order to specify the choices of the sets X and Y, some authors define a binary relation or correspondence as an ordered triple (X, Y, G), where G is a subset of $X \times Y$ called the graph of the binary relation. The statement $(x, y) \in R$ reads "x is R-related to y" and is denoted by xRy. The domain of definition or active domain of R is the set of all $x \in X$ such that there exists $y \in Y$ such that xRy. The codomain of definition, active codomain, image, or range of R is the set of all $y \in Y$ such that there exists $x \in X$ such that xRy. The field of R is the union of its domain of definition and its codomain of definition.

ii Homogeneous relation or endorelation

A homogeneous relation or endorelation on a set X is a binary relation between X and itself, i.e. it is a subset of X^2, commonly phrased as "a (binary) relation on/over X".

iii Transitive relation

A homogeneous relation R on the set X is a transitive relation if

$$\forall a, b, c \in X : (aRb \wedge bRc) \implies aRc.$$

iv Incomparability

For a homogeneous relation R on the set X, the homogeneous relation "incomparable with respect to R" is defined as "a and b in X are incomparable with respect to R if and only if $(\neg aRb) \wedge (\neg bRa)$".

v Important properties

Some important properties that a homogeneous relation R over a set X may have include: for all $a, b, c \in X$,

- Reflexivity: aRa,
- Irreflexivity or anti-reflexive: $\neq aRa$,
- Symmetry: $aRb \implies bRa$,
- Asymmetry: $aRb \implies \neg bRa$ (which implies irreflexivity),
- Antisymmetry: $aRb \wedge bRa \implies a = b$,
- Transitivity: $(aRb \wedge bRc) \implies aRc$,

- Transitivity of incomparability or comparability modulo equivalence: the homogeneous relation "imcomparable with respect to R" is transitive, that is, an equivalence relation.
- Connectivity: $a \neq b \implies (aRb \vee bRa)$ (which implies transitivity of incomparability),
- Strongly connectivity or total connectivity: $aRb \vee bRa$ (which implies reflexivity and connectivity),
- Dense property: $aRb \implies (\exists z \in X$ s.t. $aRz \wedge zRb$,
- Well-founded property: for every non-empty subset $S \subseteq X$, there exists some $m \in S$ such that mRs for all $s \in S \setminus \{m\}$.

II Equivalence relation (等價關係)

i Equivalence relation

An equivalence relation on a set X is a binary relation on X that is reflexive, symmetric, and transitive, often denoted as $\cong$, $=$, $\equiv$, or $\sim$.

ii Equivalence class (等價類)

An equivalence class of an element a in a set X under an equivalance relation $\cong$ on X is defined as

$$[a] = \{x \in X \mid a \cong x\}.$$

iii Quotient set (商集)

The quotient set $Y = X/\cong$ is the set of equivalence classes of elements of X by equivalence relation $\cong$.

III Order (序)

i Ordered set (序集)

An ordered set is an ordered pair $P = (X, \leq)$ (or $P = (X, <)$) consisting of a set X, called the ground set of P, and an order (序), aka ordering, $\leq$ (or $<$), in which an order is a transitive relation on X.

When the meaning is clear from context and there is no ambiguity about the order, the ground set itself can sometimes represents the whole ordered set.

ii Preorder (預序), quasiorder, or quasi-order

A preorder, quasiorder, or quasi-ordering is a transitive relation $\leq$ that is reflexive.

A set equipped with a preorder is called a preordered set, preset, quasiordered set, or quasi-ordered set.

iii Total preorder (全預序), total quasiorder, or total quasi-order

A total preorder is a transitive relation $\leq$ that is strongly connected.

A set equipped with a total preorder is called a totally preordered set, totally preset, totally quasiordered set, or totally quasi-ordered set.

iv Partial order (偏序)

A partial order is a transitive relation $\leq$ that is reflexive and antisymmetric.

A set equipped with a partial order is called a partially ordered set or poset.

v Upward closure

Let A be a subset of a poset X, the upward closure of A, denoted as $\uparrow A$, is defined as:

$$\uparrow A \coloneqq \{x \in X \mid \exists a \in A \text{ s.t. } a \leq x\}.$$

vi Downward closure

Let A be a subset of a poset X, the downward closure of A, denoted as $\downarrow A$, is defined as:

$$\downarrow A \coloneqq \{x \in X \mid \exists a \in A \text{ s.t. } a \geq x\}.$$

vii Closed upward/upwards

A subset A of a of a poset X is called closed upward/upwards iff

$$(x \in X \wedge y \in A \wedge x \geq y) \implies (x \in A).$$

viii Closed downward/downwards

A subset A of a of a poset X is called closed downward/downwards iff

$$(x \in X \wedge y \in A \wedge x \leq y) \implies (x \in A).$$

ix Strict partial order, strict preorder, strict quasiorder, or strict quasi-order

A strict partial order, strict preorder, strict quasiorder, or strict quasi-order is a transitive relation $<$ that is asymmetric.

A set equipped with a strict partial order is called a strictly partially ordered set, strictly preordered set, strictly preset, strictly quasiordered set, or strictly quasi-ordered set.

A strict partial order $<$ can be induced from a preorder $\leq$ with $a < b \iff (a \leq b \wedge \neg(b \leq a))$. A partial order $\leq$ can be induced from a strict partial order $<$ and an equivalence relation $=$ with $a \leq b \iff (a < b \vee a = b)$.

x Strict weak order

A strict weak order is a transitive relation $<$ that is asymmetric and with transitivity of incomparability.

A strict weak order $<$ can be induced from a preorder $\leq$ with $a < b \iff (a \leq b \wedge \neg(b \leq a)$. A partial order $\leq$ can be induced from a strict weak order $<$ with $a \leq b \iff (a < b \vee a = b)$ where $=$ denotes the equivalence relation "incomparable with respect to $<$".

xi Total order (全序) or linear order (線性順序)

A total order or linear order is a transitive relation $\leq$ that is strongly connected and antisymmetric.

A set equipped with a total order is called a totally ordered set or linearly ordered set.

xii Strict total order, strict linear order, or strict total preorder

A strict total order, strict linear order, or strict total preorder is a transitive relation $<$ that is asymmetric and connected.

A set equipped with a strict total order is called a strictly totally ordered set, strictly linearly ordered set, or strictly totally preordered set.

A strict total order $<$ can be induced from a total preorder $\leq$ with $a < b \iff (a \leq b \wedge \neg(b \leq a)$. A total order $\leq$ can be induced from a strict total order $<$ with $a \leq b \iff (a < b \vee a = b)$ where $=$ denotes the equivalence relation "incomparable with respect to $<$".

xiii (Totally/linearly) ordered group (有序群)

A group $(G, *)$ together with a total order $\leq$ on F is a

- left-(totally/linearly) ordered group if $\leq$ is left-invariant, that is

$$\forall a, b, c \in G: \; a \leq b \implies c * a \leq c * b.$$

- right-(totally/linearly) ordered group if $\leq$ is right-invariant, that is

$$\forall a, b, c \in G: \; a \leq b \implies a * c \leq b * c.$$

- bi-(totally/linearly) ordered group if $\leq$ is bi-invariant, that is it is both left- and right-invariant.

If $*$ is commutative, then left-ordred, right-ordered, and bi-ordered groups are the same, called (totally/linearly) ordered group.

xiv (Totally/linearly) ordered field (有序域)

A field $(F, +, \cdot)$ together with a total order $\leq$ on F is a (totally/linearly) ordered field if the order satisfies the following properties for all $a, b, c \in F$:

$$a \leq b \implies a + c \leq b + c,$$

$$0 \le a \wedge 0 \le b \implies 0 \le a \cdot b.$$

Elements $a \in F$ with $a > 0$ are called positive; $a \in F$ with $a < 0$ are called negative.

xv Prewellorder (預良序)

A prewellorder is a transitive relation $\le$ that is strongly connected and well-founded.

A set equipped with a prewellorder is called a prewellordered set.

xvi Well order or well-order (良序)

A well order or well-order is a transitive relation $\le$ that is strongly connected, antisymmetric, and well-founded.

A set equipped with a well-order is called a well ordered set, well-ordered set, or woset.

xvii Strict well order or strict well-order

A strict well order or strict well-order is a transitive relation $<$ that is asymmetric, connected, and well-founded.

A set equipped with a strict well order is called a strictly well ordered set or strictly well-ordered set.

A strict well order $<$ can be induced from a prewellorder $\le$ with $a < b \iff (a \le b \wedge \neg(b \le a)$. A well order $\le$ can be induced from a strict well order $<$ with $a \le b \iff (a < b \vee a = b)$ where $=$ denotes the equivalence relation ”incomparable with respect to $<$”.

xviii Dedekind-complete

A set X equipped with a total order $\le$ is Dedekind-complete if every nonempty subset S of X with an upper bound in X has a least upper bound (aka supremum) in X.

xix Order homomorphism or Order-preserving function

Given two posets $(S, \le_S)$ and $(T, \le_T)$, an order homomorphism or order-preserving function from $(S, \le_S)$ to $(T, \le_T)$ is a function f from S to T with the property that, for every $x, y \in S$, $x \le_S y$ implies $f(x) \le_T f(y)$.

xx Order isomorphism

Given two posets $(S, \le_S)$ and $(T, \le_T)$, an order isomorphism from $(S, \le_S)$ to $(T, \le_T)$ is a bijective function f from S to T with the property that, for every $x, y \in S$, $x \le_S y$ if and only if $f(x) \le_T f(y)$.

Two sets are called to be order-isomorphic if there exists an order isomorphism between them.

xxi Ordered by inclusion

A family of sets is called to be ordered by inclusion if it is a poset such that for any two sets A, B in it, $A \le B \iff A \subseteq B$.

xxii Cofinal (共尾的) or frequent

A subset $B \subseteq A$ of a preordered set $(A, \leq)$ is said to be cofinal or frequent in A if for every $a \in A$, there exists a $b \in B$ such that $a \leq b$.

xxiii Upper Bound (上界)

M is an upper bound of a preset A if $a \leq M$ for every $a \in A$. We say a preset is bounded(-)above by M if it has an upper bound M.

M is an upper bound of a function f of which the range is a preset if M is an upper bound of its range. We say a function is bounded-above by M if it has an upper bound M.

M is an upper bound of a function f of which the range is a preset on a subset I of its domain if M is an upper bound of $f(I)$. We say a function is bounded-above by M on a subset I of its domain if it has an upper bound M on I.

xxiv Lower Bound (下界)

m is an lower bound of a preset A if $a \geq m$ for every $a \in A$. We say a preset is bounded(-)below by m if it has an lower bound m.

m is an lower bound of a function f of which the range is a preset if m is an lower bound of its range. We say a function is bounded-below by m if it has an lower bound m.

m is an lower bound of a function f of which the range is a preset on a subset I of its domain if m is an lower bound of $f(I)$. We say a function is bounded-below by m on a subset I of its domain if it has an lower bound m on I.

xxv Supremum, join, or Least Upper Bound (最小上界)

The supremum, join, or least upper bound of a preset A, denoted as $\sup A$, is defined as M_0 in the set $\mathcal{M}$ of all upper bounds of A such that $\forall M \in \mathcal{M}: \ M_0 \leq M$.

The supremum or least upper bound of a function f of which the range is a preset, denoted as $\sup_x f(x)$, is defined as M_0 in the set $\mathcal{M}$ of all upper bounds of f such that $\forall M \in \mathcal{M}: \ M_0 \leq M$.

xxvi Infimum, meet, or Greatest Lower Bound (最大下界)

The infimum, meet, or greatest lower bound of a preset A, denoted as $\inf A$, is defined as m_0 in the set $\mathcal{M}$ of all lower bounds of A such that $\forall m \in \mathcal{M}: \ m_0 \geq m$.

The infimum or greatest lower bound of a function f of which the range is a preset, denoted as $\inf_x f(x)$, is defined as m_0 in the set $\mathcal{M}$ of all lower bounds of f such that $\forall m \in \mathcal{M}: \ m_0 \geq m$.

xxvii Bounded (有界的)

We say a preset or a function of which the range is a preset of which the range is a preset is bounded if it is bounded-above and bounded-below.

We say a function of which the range is a preset is bounded on a subset I of its domain if it is bounded-above and bounded-below on I.

We say a function f of which the range is a preset is unbounded at a point a in its domain if f is unbounded on all neighborhood of a.

xxviii Join-semilattice

A join-semilattice is a partially ordered set such that every two-element $\{a, b\}$ subset has a join.

xxix Meet-semilattice

A meet-semilattice is a partially ordered set such that every two-element $\{a, b\}$ subset has a meet.

xxx Lattice

A lattice is a partially ordered set such that every two-element $\{a, b\}$ subset has a join and a meet.

xxxi Directed set (有向集), directed preset (有向預序集), or filtered set

A directed set is a preset in which every finite subset has an upper bound.

xxxii Product of directed sets

Suppose $(A_i, \leq_i)_{i\in I}$ is an indexed family of directed sets. $A = \prod_{i\in I} A_i$ is a directed set equipped with coordinatewise order $(a_i)_{i\in I} \leq (b_i)_{i\in I} \equiv (a_i \leq_i b_i)_{i\in I}$.

xxxiii Intervals (區間)

Unless otherwise specified, an interval defined for $\mathbb{R}$.

For a partially ordered set $(P, \leq)$ and $a, b \in P$,

- a closed interval (閉區間) $[a, b]$ is defined as

$$[a, b] \coloneqq \{x \in P \mid a \leq x \leq b\};$$

- an left-closed upward interval $[a, \infty)$ is defined as

$$[a, \infty) \coloneqq \{x \in P \mid a \leq x\};$$

- a right-closed downward interval $(-\infty, b]$ is defined as

$$(-\infty, b] \coloneqq \{x \in P \mid x \leq b\};$$

- the open interval $(-\infty, \infty)$ is defined as

$$(-\infty, \infty) \coloneqq P.$$

For a strictly partially ordered set $(P, <)$ and $a, b \in P$,

- an open interval (開區間) (a, b) is defined as

$$(a, b) \coloneqq \{x \in P \mid a < x < b\};$$

- a right open interval (右開區間) $[a, b)$ is defined as

$$[a, b) \coloneqq \{x \in P \mid a \leq x < b\};$$

- a left open interval (左開區間) $(a, b]$ is defined as

$$(a, b] \coloneqq \{x \in P \mid a < x \leq b\};$$

- an left-open upward interval (a, ∞) is defined as

$$(a, \infty) \coloneqq \{x \in P \mid a < x\};$$

- a right-open downward interval $(-\infty, b)$ is defined as

$$(-\infty, b) \coloneqq \{x \in P \mid x < b\}.$$

Right open intervals and left open intervals are called half open intervals (半開區間). Intervals with at endpoint ∞ or $-\infty$ are called infinite intervals (無限區間) or unbounded intervals. All intervals that are not infinite intervals are called bounded intervals. The length of a bounded interval is the difference of the two endpoints of it.